

Algebraic Anomalies in ECDSA Signatures and Private Key Recovery

Formal mathematical framework derived from the implementation of `generate_transactions.py`

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§1 Base Definitions and Notation

Let an elliptic curve be defined by parameters (p, a, b, G, n) , where n is the order of the base point G . For an ECDSA signature, we introduce the following notation:

DEF 1.1

$z = H(m) \bmod n$	(message hash)
$k \in [1, n - 1]$	(ephemeral nonce)
$r = [k \cdot G]_x \bmod n$	(x-coordinate of point kG)
$s = k^{-1}(z + r \cdot x) \bmod n$	(signature scalar)
	where x is the private key

For analysis of the signature's internal structure, three *hidden components* are introduced — they are not publicly observable:

DEF 1.2 – Hidden components

$$\begin{aligned}s_{zk} &= z \cdot k^{-1} \bmod n \\ s_{rxk} &= r \cdot x \cdot k^{-1} \bmod n \\ s_{zr} &= z \cdot r \bmod n\end{aligned}$$

From the definition of the signature, the additive identity follows immediately:

EQ 1.3

$$s \equiv s_{zk} + s_{rxk} \pmod{n}$$

§2 Standard ECDSA Equation and Component Decomposition

We rewrite the classical equation in terms of the introduced components. Multiplying both sides by k^{-1} :

EQ 2.1

$$s \cdot k \equiv z + r \cdot x \pmod{n}$$

The component s_{zk} is the "projection" of the message hash onto the nonce axis, while s_{rxk} is the projection of the private key:

EQ 2.2

$$s_{zk} = \frac{z}{k} \bmod n, \quad s_{rxk} = \frac{r \cdot x}{k} \bmod n$$

From this, the private key can be expressed analytically, provided s_{zk} is known:

EQ 2.3 – Universal recovery kernel

$$x = s_{rxk} \cdot (r \cdot k)^{-1} \bmod n = (s - s_{zk}) \cdot (r \cdot s_{zk} \cdot z^{-1})^{-1} \bmod n$$

This formula is the *recovery kernel* for all anomaly classes — the difference between classes lies solely in how s_{zk} is determined.

§3 Anomaly Existence Prerequisite (Gate Condition)

Before checking classes A–D, the code computes an auxiliary value a . This is only possible when the *gate condition* is satisfied:

DEF 3.1 – Gate condition

$$s_{zr} > s \iff z \cdot r \bmod n > s$$

DEF 3.2 – Auxiliary modular remainder

$$a = s \bmod (s_{zr} - s), \quad a \in \mathbb{Z}_{>0}, \quad a \neq s, \quad a \neq s_{zr}$$

Analytically, a is the smallest non-negative remainder of dividing s by the difference $(s_{zr} - s)$. Two rational coefficients that determine the anomaly class:

DEF 3.3 – Rational anomaly coefficients

$$m_1 = \frac{s_{zk}}{a} \in \mathbb{Q}, \quad m_2 = \frac{s + s_{zr}}{s_{rxk}} \in \mathbb{Q}$$

The integrality or fractional nature of these coefficients determines the anomaly class (A, B, C, or D).

§4 Anomaly Class A — Linear Condition

DETECT

Identification Condition

CLASS A CONDITION

The gate condition holds, and additionally:

COND A.1

$$m_1 \in \mathbb{Z} \quad \wedge \quad m_1 = 1 \quad \wedge \quad m_2 \notin \mathbb{Z}$$

Since $m_1 = s_{zk}/a = 1$, the condition is equivalent to:

COND A.2

$$s_{zk} \equiv a \pmod n \iff a \mid s_{zk} \text{ and } s_{zk}/a = 1$$

RECOVER

Private Key Recovery

Step-by-step derivation from the `_recover_private_key_case_a` function with $m = 1$:

PROOF A

$$s_{zk} = a \cdot m \bmod n$$
$$k = s_{zk} \cdot z^{-1} \bmod n = a \cdot z^{-1} \bmod n$$
$$s_{rxk} = (s - s_{zk}) \bmod n$$
$$rk = r \cdot k \bmod n$$
$$x = s_{rxk} \cdot (rk)^{-1} \bmod n$$

(with m=1: $s_{zk} = a$)

CLOSED-FORM FORMULA – CASE A

FORMULA A

$$x = (s - a) \cdot z \cdot (r \cdot a)^{-1} \bmod n$$

§5 Anomaly Class B — Quadratic Condition

DETECT

Identification Condition

CLASS B CONDITION

$$m_1 \in \mathbb{Z} \quad \wedge \quad m_2 \in \mathbb{Z} \quad \wedge \quad m_1 = m_2$$

That is, both rational coefficients are integers and equal to each other. Expanded, this means:

$$\frac{s_{zk}}{a} = \frac{s + s_{zr}}{s_{rxk}} = m \in \mathbb{Z}$$

Since $s = s_{zk} + s_{rxk}$ and $s_{zk} = am$, substitution gives:

$$am + s_{rxk} = s, \quad s_{rxk}(m) = s - am$$

The equation $\frac{s+s_{zr}}{s_{rxk}} = m$ transforms into a quadratic in m :

$$a \cdot m^2 - s \cdot m + (s + s_{zr}) \equiv 0 \pmod{n}$$

RECOVER

Private Key Recovery

Solving the quadratic equation via discriminant modulo n :

$$\Delta = (s^2 - 4a \cdot (s_{zr} + s)) \pmod{n}$$

$$\sqrt{\Delta} \pmod{n} \Rightarrow \text{all roots } \delta_i \text{ such that } \delta_i^2 \equiv \Delta \pmod{n}$$

$$m_i = (s + \delta_i) \cdot (2a)^{-1} \pmod{n}$$

$$x_i = (s - a \cdot m_i) \cdot z \cdot (r \cdot a \cdot m_i)^{-1} \pmod{n}$$

NOTE

For each root δ_i of the discriminant, a separate private key candidate x_i is obtained. The correct candidate is verified via the public key $P = x \cdot G$.

§6 Anomaly Class C — Intermediate Class

DETECT Identification Condition

Class C combines two sub-cases:

SUB-CASE C.1

COND C.1

$$m_1 \in \mathbb{Z} \quad \wedge \quad m_2 \in \mathbb{Z} \quad \wedge \quad m_1 \neq m_2$$

SUB-CASE C.2

COND C.2

$$m_1 \in \mathbb{Z} \quad \wedge \quad m_1 > 1 \quad \wedge \quad m_2 \notin \mathbb{Z}$$

STATUS

For class C, an analytical key recovery formula has **not yet been derived** (`x_recovered = "-"` in the transaction table).
The class is intermediate between A (where $m_1 = 1$) and B (where $m_1 = m_2$).

§7 Anomaly Class D — Fractional Condition

CLASS D CONDITION

COND D.1

$$m_1 = \frac{s_{zk}}{a} \notin \mathbb{Z} \iff a \nmid s_{zk}$$

The sub-class is determined by the integer part of the fractional ratio:

DEF D.2 – D-level

$$f = \frac{s_{zk}}{a} \in \mathbb{Q} \setminus \mathbb{Z}, \quad d = \lfloor f \rfloor \in \mathbb{Z}_{\geq 0}$$

Class: D_d , e.g. $D_0, D_1, D_2, \dots$

The fractional part $\{f\} = f - d$ carries information about the distance between s_{zk} and the nearest multiple of a :

EQ D.3 – Fractional decomposition

$$\begin{aligned} s_{zk} &= a \cdot d + a \cdot \{f\}, \quad 0 < \{f\} < 1 \\ \implies s_{zk} &= a \cdot d + \rho, \quad \rho = s_{zk} \bmod a, \quad 0 < \rho < a \end{aligned}$$

Candidate Space for D_d

Since $s_{zk} = z \cdot k^{-1} \bmod n$, candidates for k at a fixed level d satisfy:

EQ D.4 – Candidate space

$$\begin{aligned} z \cdot k^{-1} &\equiv a \cdot d + \rho \pmod{n}, \quad \rho \in (0, a) \\ \implies k &\equiv z \cdot (ad + \rho)^{-1} \pmod{n} \end{aligned}$$

ACTIVE RESEARCH STATUS

The search space for D_d is bounded by a candidates (over $\rho \in [1, a - 1]$). The method of narrowing through intersection of candidate sets x across multiple transactions of the same key is under development and validation.

§8 Anomaly Class E — Goldbach Condition

Class E is checked *independently* of the gate condition in §3 and is based on the additive symmetry of the components relative to s .

DETECT Sub-class E1

E1 - STANDARD CASE

COND E1

$$s_{zk} + s_{rxk} \leq s \quad \wedge \quad s \equiv 0 \pmod{2}$$
$$\wedge \quad |s_{zk} - s_{rxk}| = \frac{s}{2} + 1$$

DETECT Sub-class E2

E2 - WRAP-AROUND CASE

COND E2

$$s_{zk} + s_{rxk} > s \quad \Rightarrow \quad s' = s + n$$
$$s' \equiv 0 \pmod{2} \quad \wedge \quad |s_{zk} - s_{rxk}| = \frac{s'}{2} + 1$$

A single unified condition for both sub-classes (where s^* is the adjusted s):

COND E - Goldbach decomposition

$$\exists \nu = \frac{s^*}{2} \in \mathbb{Z} : \quad |s_{zk} - s_{rxk}| = \nu + 1$$

This means s_{zk} and s_{rxk} are two numbers that differ from the "midpoint" ν by exactly $\frac{1}{2}$, i.e.:

$$\{s_{zk}, s_{rxk}\} = \left\{ \nu + \frac{1}{2} + \frac{1}{2}, \nu - \frac{1}{2} + \frac{1}{2} \right\} = \{\nu + 1, \nu\} \implies s_{zk} + s_{rxk} = 2\nu + 1 = s^* \pm 1$$

RECOVER

Private Key Recovery

From s^* we compute two candidates a and b for the component s_{zk} :

FORMULA E.1 – Candidate components

$$\nu = \left\lfloor \frac{s^*}{2} \right\rfloor, \quad a = \left\lceil \frac{s^* + \nu + 1}{2} \right\rceil = \frac{s^* + \nu + 1}{2}, \quad b = s^* - a$$

Each candidate yields a separate private key via the recovery formula from §2:

FORMULA E.2 – Dual key recovery

$$x_1 = (s^* - a) \cdot z \cdot (r \cdot a)^{-1} \bmod n$$

$$x_2 = (s^* - b) \cdot z \cdot (r \cdot b)^{-1} \bmod n$$

CONNECTION TO GOLDBACH'S CONJECTURE

Condition E is a discrete analogue of decomposing an even number s^* into a sum of two "almost equal" addends. The asymmetry of ± 1 relative to the midpoint ν is the key signature of this class.

§9 Summary Table of Anomaly Classes

CLASS	DETECTION CONDITION	RECOVERY FORMULA FOR X	STATUS
A	$m_1 \in \mathbb{Z}, m_1 = 1, m_2 \notin \mathbb{Z}$	$x = (s - a) \cdot z \cdot (ra)^{-1} \bmod n$	✓ Derived
B	$m_1 \in \mathbb{Z}, m_2 \in \mathbb{Z}, m_1 = m_2$	$x_i = (s - am_i) \cdot z \cdot (ram_i)^{-1} \bmod n$ where m_i are roots of the quadratic equation	✓ Derived

CLASS	DETECTION CONDITION	RECOVERY FORMULA FOR X	STATUS
<code>C</code>	$(m_1 \in \mathbb{Z}, m_2 \in \mathbb{Z}, m_1 \neq m_2)$ or $(m_1 \in \mathbb{Z}, m_1 > 1, m_2 \notin \mathbb{Z})$	—	X Not derived
<code>D\(_d\)</code>	$m_1 \notin \mathbb{Z}, d = \lfloor m_1 \rfloor$	a candidates; search via intersection	🔗 Active research
<code>E1/E2</code>	$ s_{zk} - s_{rxk} = \lfloor s^*/2 \rfloor + 1$	$x_{1,2} = (s^* - c) \cdot z \cdot (rc)^{-1} \bmod n$ for $c \in \{a, b\}$	✓ Derived